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(54) **SERVER**

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(57) **ABSTRACT**

The server refers to the predicted moving body information to identify the first group and the second group. The server calculates the influence degree for each group. The server calculates a display state of the traffic rule display device in which the highly influence degree group among the first group and the second group can continue traveling. The server transmits the calculated request to be in the display state.

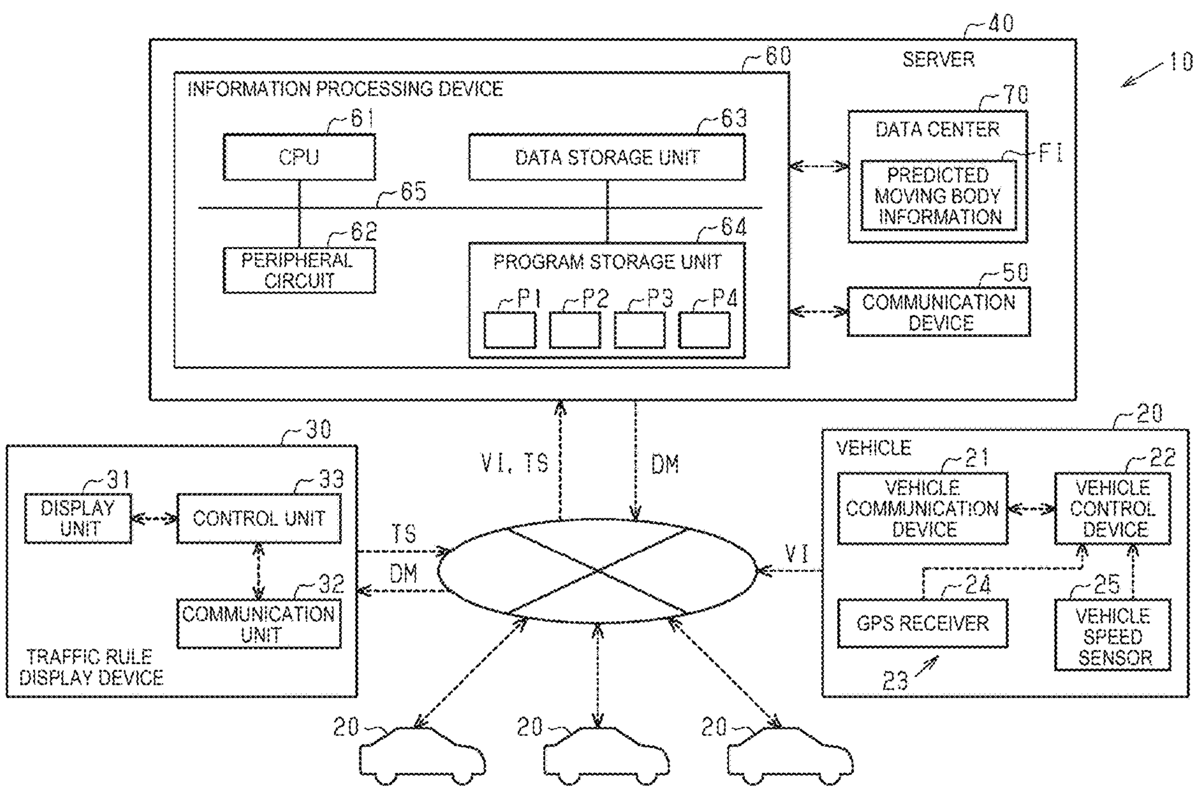


FIG. 1

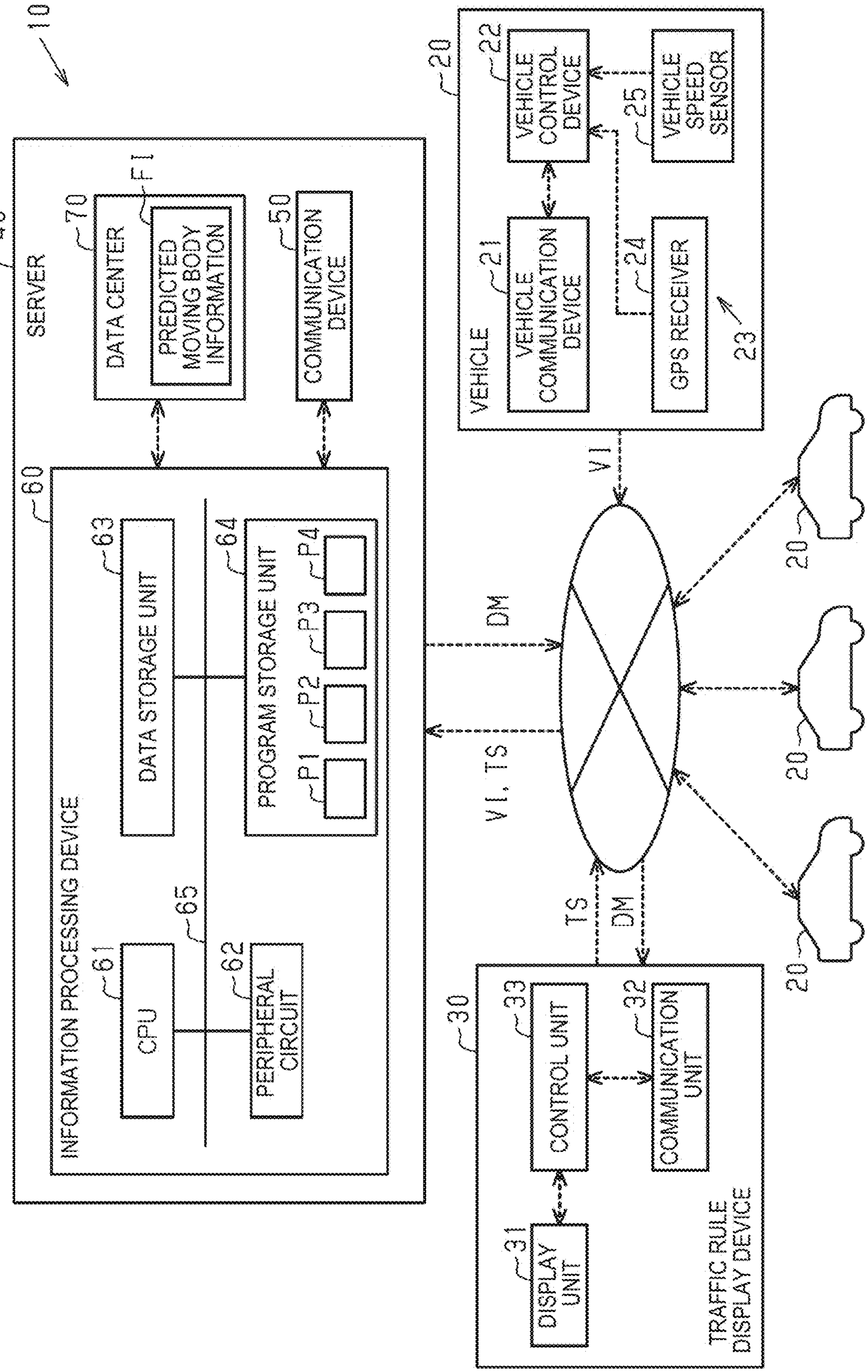

10
SERVER
40
DATA CENTER
70
PREDICTED MOVING BODY INFORMATION
FI
COMMUNICATION DEVICE
50
INFORMATION PROCESSING DEVICE
60
CPU
61
PERIPHERAL CIRCUIT
62
DATA STORAGE UNIT
63
PROGRAM STORAGE UNIT
64
65
P1
P2
P3
P4
VEHICLE
20
VEHICLE COMMUNICATION DEVICE
21
VEHICLE CONTROL DEVICE
22
GPS RECEIVER
24
23
VEHICLE SPEED SENSOR
25
VI
DM
VI, TS
TS
DM
TRAFFIC RULE DISPLAY DEVICE
30
DISPLAY UNIT
31
COMMUNICATION UNIT
32
CONTROL UNIT
33
20
20
20

FIG. 2

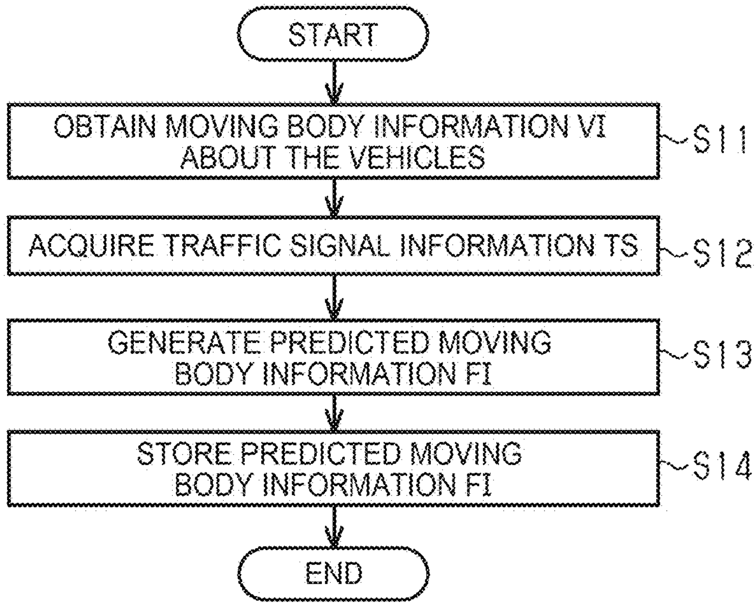

START
OBTAIN MOVING BODY INFORMATION VI ABOUT THE VEHICLES
S11
ACQUIRE TRAFFIC SIGNAL INFORMATION TS
S12
GENERATE PREDICTED MOVING BODY INFORMATION FI
S13
STORE PREDICTED MOVING BODY INFORMATION FI
S14
END

FIG. 3

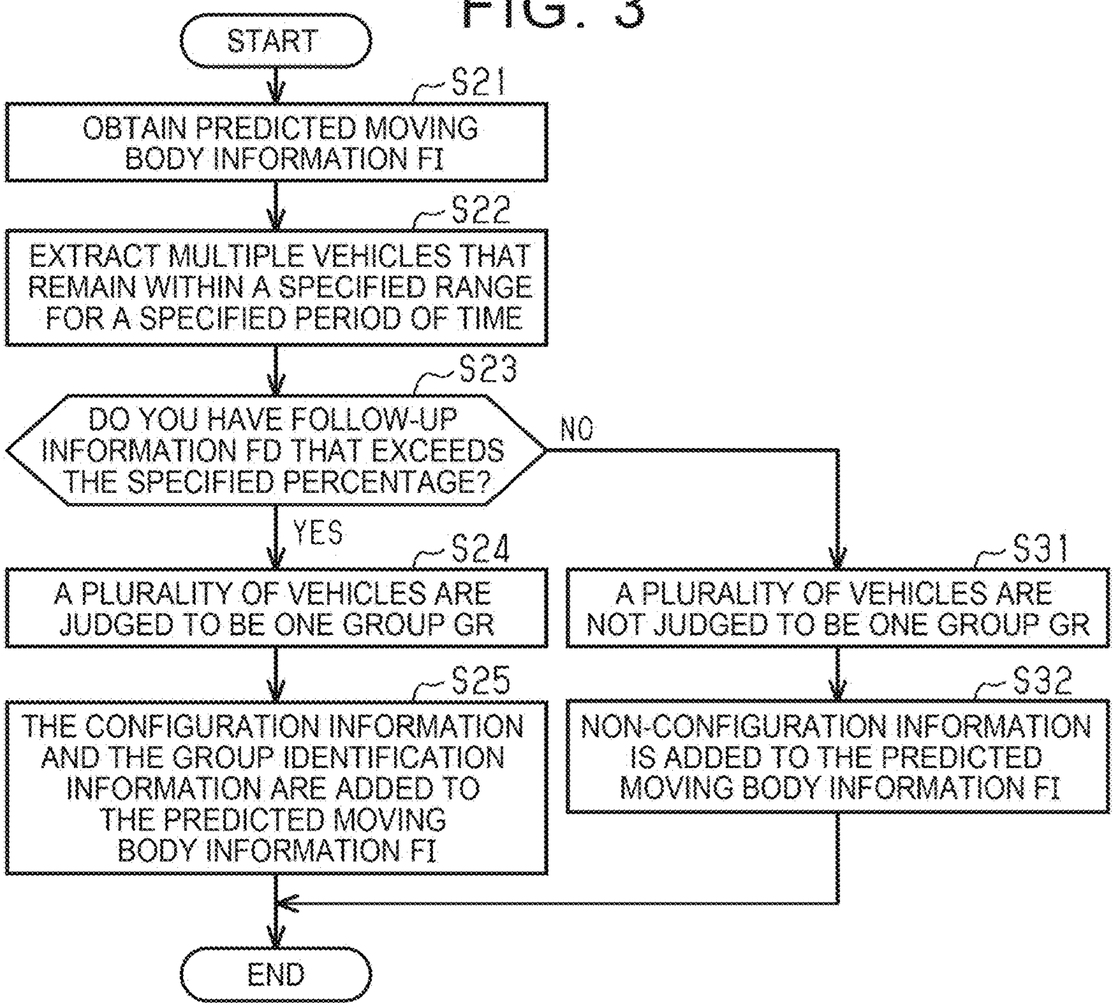

START
S21
OBTAIN PREDICTED MOVING BODY INFORMATION FI
S22
EXTRACT MULTIPLE VEHICLES THAT REMAIN WITHIN A SPECIFIED RANGE FOR A SPECIFIED PERIOD OF TIME
S23
DO YOU HAVE FOLLOW-UP INFORMATION FD THAT EXCEEDS THE SPECIFIED PERCENTAGE?
NO
YES
S24
A PLURALITY OF VEHICLES ARE JUDGED TO BE ONE GROUP GR
S31
A PLURALITY OF VEHICLES ARE NOT JUDGED TO BE ONE GROUP GR
S25
THE CONFIGURATION INFORMATION AND THE GROUP IDENTIFICATION INFORMATION ARE ADDED TO THE PREDICTED MOVING BODY INFORMATION FI
S32
NON-CONFIGURATION INFORMATION IS ADDED TO THE PREDICTED MOVING BODY INFORMATION FI
END

FIG. 4

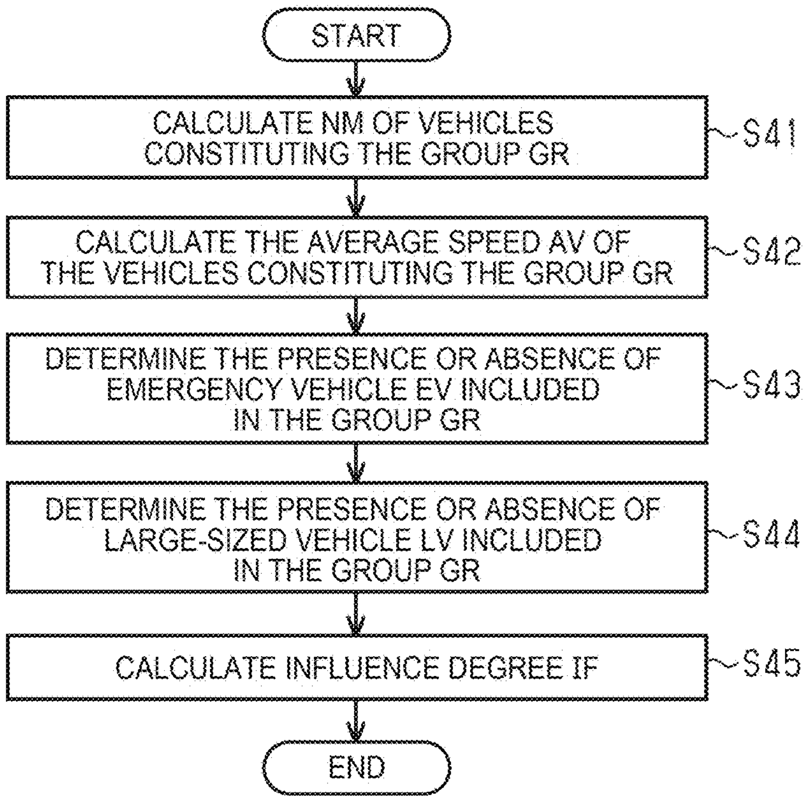
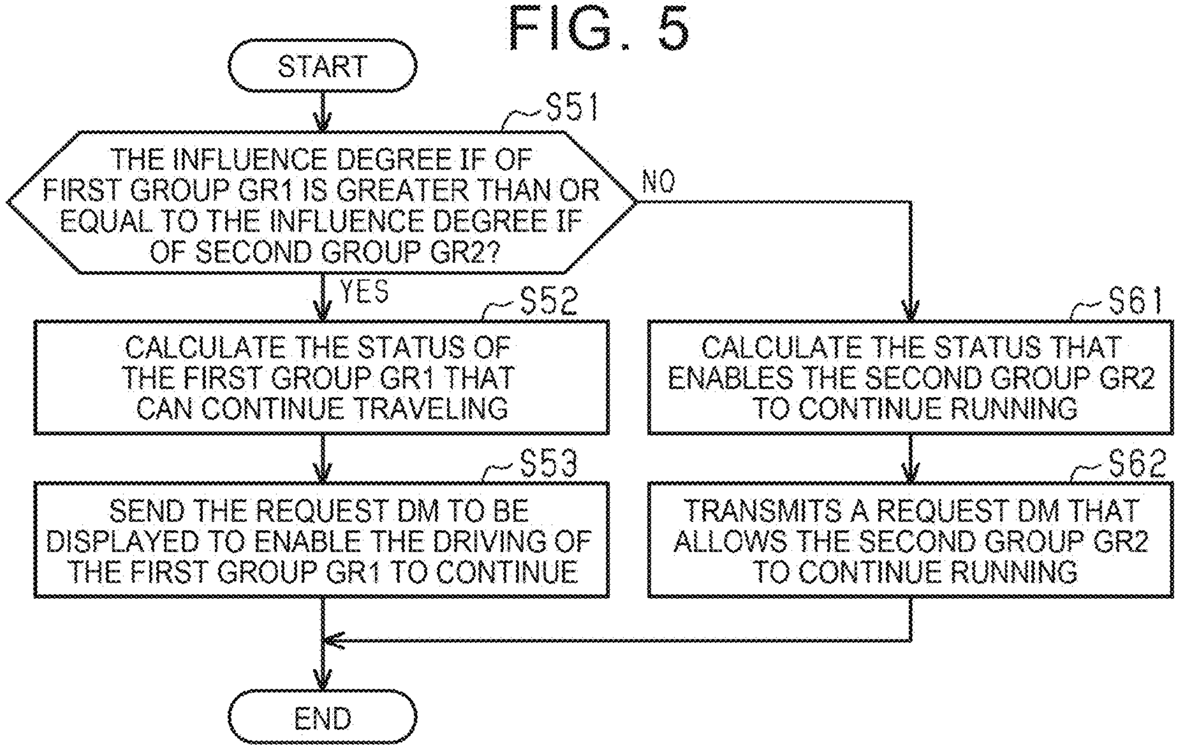

START
CALCULATE NM OF VEHICLES CONSTITUTING THE GROUP GR
S41
CALCULATE THE AVERAGE SPEED AV OF THE VEHICLES CONSTITUTING THE GROUP GR
S42
DETERMINE THE PRESENCE OR ABSENCE OF EMERGENCY VEHICLE EV INCLUDED IN THE GROUP GR
S43
DETERMINE THE PRESENCE OR ABSENCE OF LARGE-SIZED VEHICLE LV INCLUDED IN THE GROUP GR
S44
CALCULATE INFLUENCE DEGREE IF
S45
END

FIG. 5


START
S51
THE INFLUENCE DEGREE IF OF FIRST GROUP GR1 IS GREATER THAN OR EQUAL TO THE INFLUENCE DEGREE IF OF SECOND GROUP GR2?
NO
YES
S52
CALCULATE THE STATUS OF THE FIRST GROUP GR1 THAT CAN CONTINUE TRAVELING
S61
CALCULATE THE STATUS THAT ENABLES THE SECOND GROUP GR2 TO CONTINUE RUNNING
S53
SEND THE REQUEST DM TO BE DISPLAYED TO ENABLE THE DRIVING OF THE FIRST GROUP GR1 TO CONTINUE
S62
TRANSMITS A REQUEST DM THAT ALLOWS THE SECOND GROUP GR2 TO CONTINUE RUNNING
END

SERVER

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-036746 filed on Mar. 11, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The disclosure relates to a server.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2017-167669 (JP 2017-167669 A) describes a communication system including a vehicle and a server. When a plurality of vehicles travels as a group, the server transmits a control signal to the vehicles traveling in the group.

SUMMARY

[0004] Now, in a traffic network including a plurality of vehicles, a plurality of groups may pass by a traffic rule display device such as a traffic signal. In a traffic network including a communication system as described in JP 2017-167669 A, there is room for improving efficiency of the traffic network as a whole.

[0005] To solve the above problems, the disclosure is a server that transmits a control signal based on predicted moving body information, that is generated based on moving body information that is information of a vehicle in a real world, following a point in time of acquiring the moving body information, to a traffic rule display device that displays different traffic rules by changing a display state, the server executing identifying a first group, and a second group different from the first group, as groups in which a plurality of the vehicles travels in a group, based on the predicted moving body information, calculating, for each of the groups identified, an influence degree, indicating a degree of influence imparted on a traffic network when the group that is an object of calculation is stopped, calculating, when the first group and the second group are located within a vicinity range including a range of traveling in accordance with a display of the traffic rule display device, the display state under which the group of the first group and the second group of which the influence degree is higher can continue traveling, and transmitting a request, that is the display state that is calculated, to the traffic rule display device.

[0006] The server transmits the request to the traffic rule display device, thereby enabling a group, of which the influence degree is high among the first group and the second group, to continue traveling. Thus, the server can suppress influence imparted on the traffic network due to the group having a high influence degree stopping.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0008] FIG. **1** is a schematic diagram illustrating a communication system;

[0009] FIG. **2** is a flowchart illustrating a series of processes for generating predicted moving body information;

[0010] FIG. **3** is a flowchart illustrating a series of processes for determining a group;

[0011] FIG. **4** is a flow chart showing a series of processes for calculating the influence degree; and

[0012] FIG. **5** is a flowchart illustrating a series of processes for transmitting a request to a traffic rule display device.

DETAILED DESCRIPTION OF EMBODIMENTS

Embodiment

[0013] Hereinafter, an embodiment of a server will be described with reference to the drawings. Hereinafter, a communication system including a server will be described.

Outline of the Communication System

[0014] As illustrated in FIG. **1**, the communication system **10** includes a plurality of vehicles **20**, a traffic rule display device **30**, and a server **40**.

[0015] The vehicle **20** includes a vehicle communication device **21**, a vehicle control device **22**, and a plurality of information acquisition devices **23**. The vehicle communication device **21** communicates with the server **40** by wireless communication. The vehicle control device **22** controls communication of the vehicle communication device **21**.

[0016] The plurality of information acquisition devices **23** acquire various types of information of the vehicle **20**. The plurality of information acquisition devices **23** are, for example, a GPS receiving device **24** and a vehicle speed sensor **25**. GPS receiving device **24** receives position data indicating the position of the vehicles **20** from GPS device. The position information is a coordinate value of latitude and longitude. The vehicle speed sensor **25** acquires the traveling speed of the vehicle **20** as the vehicle speed. Each information acquisition device **23** outputs the acquired information of the vehicle **20** to the vehicle control device **22**.

[0017] The vehicle control device **22** controls the following travel of the vehicle **20**. The vehicle control device **22** controls, by a user's operation, the following travel that follows the other vehicle **20** traveling in front. The vehicle control device **22** generates a follow-up information FD indicating that the vehicle is in follow-up travel when the vehicle is in follow-up travel.

[0018] The vehicle control device **22** acquires, as the moving body information VI, various kinds of information of the vehicle **20** acquired from the plurality of information acquisition devices **23**, identification information indicating the vehicle **20**, a point in time at which the various kinds of information are acquired, and a follow-up information FD of the vehicle **20**. The moving body information VI is information of the real-world vehicles **20**. The identification information indicating the vehicle **20** includes information indicating whether or not the vehicle **20** is in the large-sized vehicle LV and whether or not the vehicle is in the emergency vehicle EV. The large-sized vehicle LV is a vehicle **20** having a total weight of a predetermined weight or more, or a vehicle **20** having a number of persons or more determined in advance by the occupant capacity. Large-sized vehicles

LV are large trucks, route buses, and the like. The emergency vehicle EV is, for example, an ambulance.

[0019] The vehicle control device **22** outputs the moving body information VI to the vehicle communication device **21**. Then, the vehicle communication device **21** transmits the moving body information VI to the servers **40**. In FIG. **1**, one vehicle **20** of the plurality of vehicles **20** is illustrated in detail, and the other vehicle **20** is illustrated in detail without detail. The vehicles **20** each transmit a moving body information VI to the servers **40**.

[0020] The traffic rule display device **30** displays a traffic rule. The traffic rule display device **30** can display different traffic rules by changing the display state. The traffic rule display device **30** is, for example, a traffic signal provided at a cross-shaped intersection.

[0021] The traffic rule display device **30** includes a display unit **31**, a communication unit **32**, and a control unit **33**. The display unit **31** includes a plurality of lamps. The plurality of lamps includes a red lamp indicating that entry is prohibited at an intersection of a range traveling in accordance with the display of the traffic rule display device **30**, and a blue lamp indicating that entry is possible at the intersection.

[0022] The communication unit **32** communicates with the server **40** by wireless communication. The communication unit **32** transmits a signal indicating the display state of the display unit **31** to the server **40**. The communication unit **32** receives a signal for controlling the display unit **31** from the server **40**.

[0023] The control unit **33** controls the display state of the display unit **31**. For example, the control unit **33** controls a state in which the red lamp is not turned on and the blue lamp is turned on. Further, the control unit **33** controls the display state of the display unit **31** based on a signal for controlling the display unit **31** received by the communication unit **32**. The control unit **33** acquires information indicating the display state of the display unit **31**, identification information indicating the traffic rule display device **30**, and information indicating the point in time in the display state as traffic signal information TS. Then, the control unit **33** outputs the traffic signal information TS to the communication unit **32**. The communication unit **32** then transmits the traffic signal information TS to the servers **40**.

[0024] The server **40** is capable of communicating with a plurality of vehicles **20**. In addition, the server **40** can communicate with the traffic rule display device **30**. The servers **40** acquire the plurality of moving body information VI from the plurality of vehicles **20**. The servers **40** acquire the traffic signal information TS from the traffic rule display device **30**. The server **40** can transmit a control signal based on the predicted moving body information FI generated based on the moving body information VI, which will be described later, to the traffic rule display device **30**. The server **40** includes a communication device **50**, an information processing device **60**, and a data center **70**.

[0025] The communication device **50** communicates with a plurality of vehicles **20**. The communication device **50** receives the moving body information VI transmitted from the vehicles **20**. The communication device **50** receives the traffic signal information TS transmitted from the traffic rule display device **30**. The communication device **50** outputs the received moving body information VI and the received traffic signal information TS to the information processing device **60**. Further, the communication device **50** transmits the information acquired from the information processing device **60** to the vehicle **20**. The communication device **50** transmits the signal acquired from the information processing device **60** to the traffic rule display device **30**.

[0026] The information processing device **60** includes a CPU **61** as an executing device, peripheral circuit **62**, a data storage unit **63**, a program storage unit **64**, and a bus **65**. The bus **65** communicatively connects CPU **61**, the peripheral circuit **62**, the data-storage unit **63**, and the program-storage unit **64** to each other. The peripheral circuit **62** includes a circuit that generates a clock signal that defines an internal operation, a power supply circuit, a reset circuit, and the like. The data storage unit **63** stores data generated in association with the operation of CPU **61**. The program storage unit **64** stores a generation program P**1** of the predicted moving body information FI, a determination program P**2** of the group GR, a calculation program P**3** of the influence degree IF, and a control program P**4** of the traffic rule display device **30**. CPU **61** performs information processing by executing various programs stored in the program storage unit **64**.

[0027] The data center **70** stores the predicted moving body information FI. The predicted moving body information FI is information including a plurality of moving body information VI generated based on the moving body information VI of the plurality of vehicles **20** and after the point in time when the moving body information VI in the predetermined area is acquired. The predetermined area may be, for example, a range including one country, a range including only a part of a country, or a range including the last time. That is, the predicted moving body information FI is a so-called digital twin. Specifically, the data center **70** stores time-series data of the predicted moving body information FI generated by the information processing device **60**. The data center **70** acquires the predicted moving body information FI generated by the information processing device **60** a plurality of times over time. As a result, the data center **70** stores the time-series data of the predicted moving body information FI.

Generation of Predicted Moving Body Information

[0028] Next, generation of the predicted moving body information FI performed by the information processing device **60** will be described.

CPU **61** repeatedly generates the predicted moving body information FI by repeating the generation program P**1** of the predicted moving body information FI at a predetermined cycle. The predetermined period is defined as, for example, one minute.

[0029] As illustrated in FIG. **2**, when CPU **61** starts executing the program P**1** for generating the predicted moving body information FI, it first performs a S**11** process. In S**11**, CPU **61** acquires moving body information VI of the vehicles **20** in the communication system **10**. When acquiring the plurality of pieces of moving body information VI, CPU **61** acquires the moving body information VI of the respective vehicles **20** based on the identification information of the vehicles **20** included in the moving body information VI. Thereafter, CPU **61** advances the process to S**12**.

[0030] In S**12**, CPU **61** obtains the traffic signal information TS of the traffic rule display device **30**. Thereafter, CPU **61** advances the process to S**13**.

In S**13**, CPU **61** generates the predicted moving body information FI based on the plurality of acquired moving body information VI and the traffic signal information TS. Specifically, CPU **61** synchronizes the moving body information VI and the traffic signal information TS of the respective vehicles **20** by performing the following processes, so that CPU **61** generates the predicted moving body information FI. First, CPU **61** refers to information indicating the acquired point in time for the plurality of acquired moving body information VI. Next, CPU **61** predicts the moving body information VI at the reference point in time by correcting the other moving body information VI by the difference of the point in times using the point in time of the moving body information VI having the newest acquired point in time as the reference point in time. For example, CPU **61** is corrected based on a moving body information VI such as a previous vehicle speed. Next, CPU **61** predicts the display status of the traffic rule display device **30** at the reference point in time based on the acquired traffic signal information TS.

[0031] Then, CPU **61** generates various types of information of the predicted moving body information VI and the display status of the predicted traffic signal information TS as the predicted moving body information FI. As a result, CPU **61** acquires the moving body information VI and the traffic signal information TS of the plurality of vehicles **20** synchronized with the reference point in time as the predicted moving body information FI. Thereafter, CPU **61** advances the process to S**14**.

[0032] In S**14**, CPU **61** stores the generated predicted moving body information FI in the data center **70**. Thereafter, CPU **61** ends the series of processes. Accordingly, the data center **70** stores the acquired predicted moving body information FI. The data center **70** acquires and stores the predicted moving body information FI at predetermined intervals by CPU **61** repeating the generation program P**1** of the predicted moving body information FI. Therefore, the data center **70** stores the time-series data of the predicted moving body information FI.

Judgment of the Group

[0033] Next, the determination of the group GR of the group traveling performed by the information processing device **60** will be described.

[0034] CPU **61** repeats the determination-program P**2** of the group GR at a predetermined cycle. The predetermined period is defined as, for example, one minute. Accordingly, CPU **61** determines the vehicle **20** constituting the group GR and the vehicle **20** not constituting the group GR among the plurality of vehicles **20** in the predetermined area.

[0035] As shown in FIG. **3**, when CPU **61** starts executing P**2** of the determination program of the group GR, it starts S**21** process. In S**21**, CPU **61** acquires data of the time-series data of the predicted moving body information FI in the data center **70** for a predetermined time period. The past predetermined period is, for example, 10 minutes. Thereafter, CPU **61** advances the process to S**22**.

[0036] In S**22**, CPU **61** extracts a plurality of vehicles **20** that continue to exist within a predetermined range for a predetermined period in the past, based on the time-series data of the predicted moving body information FI acquired by S**21** for a predetermined period in the past. The specified range is, for example, a range in which the distance between the plurality of vehicles **20** is within 100 meters. Thereafter, CPU **61** advances the process to S**23**. In S**22**, when CPU **61** cannot extract the plurality of vehicles **20**, CPU **61** adds non-configuration information indicating that the group GR is not configured to the predicted moving body information FI for all the vehicles **20**, and ends the series of processes.

[0037] In S**23**, CPU **61** determines whether or not the number of vehicles **20** having a specified ratio or more among the plurality of vehicles **20** extracted by S**22** is following. For example, the specified ratio is set at 80%. Specifically, CPU **61** determines whether or not the follow-up information FD is included in the moving body information VI of the plurality of vehicles **20** extracted by S**22**. Then, CPU **61** compares the number of moving body information VI including the follow-up information FD with the number extracted by S**22**.

[0038] When the follow-up information FD is equal to or larger than the specified ratio (S**23**: YES), CPU **61** advances the process to S**24**. In S**24**, CPU **61** determines the plurality of vehicles **20** extracted by S**22** as a group GR traveling in one group. Thereafter, CPU **61** advances the process to S**25**.

[0039] In S**25**, CPU **61** adds configuration information indicating that the group GR is configured and group identification information identifying the configured group GR to the predicted moving body information FI for the plurality of vehicles **20** determined to be one group GR in S**24**. Thereafter, CPU **61** ends the series of processes.

[0040] On the other hand, when the follow-up information FD is not equal to or larger than the specified ratio (S**23**: YES), CPU **61** advances the process to S**31**. In S**31**, CPU **61** does not determine the plurality of vehicles **20** extracted by S**22** as one group. Thereafter, CPU **61** advances the process to S**32**.

[0041] In S**32**, CPU **61** adds, to the predicted moving body information FI, non-configuration information indicating that the group GR is not configured for the plurality of vehicles **20** that have not been determined to be one group GR in S**31**. Thereafter, CPU **61** ends the series of processes. In this way, the group GR determination program P**2** is executed so that the predicted moving body information FI includes information indicating whether or not the group GR is configured.

Calculation of Influence Degree

[0042] Next, the calculation of the influence degree IF of the group GR performed by the information processing device **60** will be described.

[0043] CPU **61** repeats the calculation program P**3** of the influence degree IF at a predetermined cycle. The predetermined period is defined as, for example, one minute. The influence degree IF indicates a degree of influence on the traffic network in a predetermined area when the group GR to be calculated is stopped. Thus, CPU **61** calculates the influence degree IF for each group GR configured in a predetermined area.

[0044] Specifically, CPU **61** refers to the predicted moving body information FI and identifies a second group GR**2** that differs from the first group GR**1** and the first group GR**1** among the plurality of group GR. Specifically, CPU **61** identifies the first group GR**1** and the second group GR**2** by referring to the information identifying the group GR included in the predicted moving body information FI. CPU **61** calculates the influence degree IF of the first group GR**1** and the influence degree IF of the second group GR**2**, respectively.

[0045] As illustrated in FIG. **4**, CPU **61** starts executing the calculation program P**3** of the influence degree IF, and CPU **61** first starts S**41** process. In S**41**, CPU **61** calculates several NM of the vehicles **20** constituting the group GR to be calculated. Specifically, CPU **61** refers to the predicted moving body information FI and calculates NM of the vehicles **20** having the information for identifying the group GR to be calculated. Thereafter, CPU **61** advances the process to S**42**.

[0046] In S**42**, CPU **61** calculates the average speed AV of the vehicles **20** included in the group GR. Specifically, CPU **61** acquires the velocity of the vehicle **20** having the information for identifying the group GR to be calculated by referring to the predicted moving body information FI of the vehicle **20** included in the group GR to be calculated. Next, CPU **61** calculates an average of the acquired speeds of the plurality of vehicles **20** as an average speed AV of the vehicles **20** included in the group GR. Thereafter, CPU **61** advances the process to S**43**.

[0047] In S**43**, CPU **61** determines whether or not there is an emergency vehicle EV included in the calculation target group GR. Specifically, CPU **61** determines whether or not various kinds of information of the predicted moving body information FI of the vehicle **20** included in the group GR include information indicating that the vehicle is in an emergency vehicle EV. Thereafter, CPU **61** advances the process to S**44**.

[0048] In S**44**, CPU **61** calculates whether or not there is a large-sized vehicle LV in GR to be calculated. CPU **61** determines whether or not various kinds of information of the predicted moving body information FI of the vehicle **20** included in the group GR include information indicating that the vehicle is a large-sized vehicle LV. Thereafter, CPU **61** advances the process to S**45**.

[0049] In S**45**, CPU **61** calculates the influence degree IF of the group GR to be calculated. Specifically, CPU **61** calculates the influence degree IF larger as the number of vehicles **20** constituting the group GR to be calculated increases. CPU **61** calculates the influence degree IF to be larger as the average speed AV of the vehicles **20** constituting the calculation target group GR is larger. When the emergency vehicle EV is included in the group GR to be calculated, CPU **61** calculates the influence degree IF larger than when the emergency vehicle EV is not included. When the large-sized vehicle LV is included in the group GR to be calculated, CPU **61** calculates the influence degree IF larger than when the large-sized vehicle LV is not included. Thereafter, CPU **61** adds information indicating the influence degree IF of the group GR to the predicted moving body information FI, and ends the series of processes.

Control of Traffic Rule Display Device

[0050] Next, the control of the traffic rule display device **30** based on the influence degree IF of the group GR performed by the information processing device **60** will be described.

[0051] When the first group GR**1** and the second group GR**2** are located within a vicinity range including a range traveling in accordance with the display of the traffic rule display device **30**, CPU **61** starts executing the control program P**4** of the traffic rule display device **30**. The vicinity range includes a range in which the vehicle travels in accordance with the display of the traffic rule display device **30**, and is determined in advance as a range wider than the range. The vicinity range includes, for example, a range of an intersection where the traffic rule display device **30** is provided, and is defined as a range of 100 meters in radius from the center of the intersection.

[0052] As illustrated in FIG. **5**, when CPU **61** starts executing the control program P**4** of the traffic rule display device **30**, it first performs S**51** process. In S**51**, CPU **61** determines whether or not the influence degree IF of the first group GR**1** is equal to or greater than the influence degree IF of the second group GR**2**. Specifically, CPU **61** refers to the predicted moving body information FI and acquires the influence degree IF of the first group GR**1** and the influence degree IF of the second group GR**2**. CPU **61** then compares the acquired influence degree IF.

[0053] When the influence degree IF of the first group GR**1** is equal to or greater than the influence degree IF of the second group GR**2** (S**51**: YES), CPU **61** advances the process to S**52**. In S**52**, CPU **61** calculates an indication that allows the first group GR**1** to continue traveling. Thereafter, CPU **61** advances the process to S**53**.

[0054] In S**53**, CPU **61** transmits, to the traffic rule display device **30**, a request DM that is a display status enabling the driving of the first group GR**1** to be continued. Thereafter, CPU **61** ends the series of processes.

[0055] On the other hand, when the influence degree IF of the first group GR**1** is less than the influence degree IF of the second group GR**2** (S**51**: NO), CPU **61** advances the process to S**61**. In S**61**, CPU **61** calculates a displaying condition in which the traveling of the second group GR**2** can be continued. Thereafter, CPU **61** advances the process to S**62**.

[0056] In S**62**, CPU **61** transmits a control signal to the traffic rule display device **30**, the control signal indicating a request DM in which the second-group GR**2** can continue traveling. Thereafter, CPU **61** ends the series of processes. In this way, CPU **61** calculates a display state in which the group GR having the higher influence degree IF among the first group GR**1** and the second group GR**2** can continue traveling, and transmits the request DM that becomes the display state to the traffic rule display device **30**.

Operations of Embodiment

[0057] Here, it is assumed that the first group GR**1** is traveling on a road traveling in the first direction at an intersection where the traffic rule display device **30** is provided. It is assumed that the second group GR**2** is traveling on a road traveling in a second direction intersecting the road traveling in the first direction at an intersection where the traffic rule display device **30** is provided. Then, when the first group GR**1** and the second group GR**2** are located in the vicinity range including the range in which the vehicle travels according to the display of the traffic rule display device **30**, CPU **61** starts executing the control-program P**4** of the traffic rule display device **30**. The traffic rule display device **30** receives the request DM indicating the state that can continue traveling of the first group GR**1** when the influence degree IF of the first group GR**1** and the influence degree IF of the second group GR**2** is large. In the traffic rule display device **30** that has received the request DM, the control unit **33** controls the display unit **31** to turn on the blue light to the road that travels in the first direction at the installed intersection. In addition, in the traffic rule display device **30**, the red lamp is turned on the road traveling in the second direction.

Effects of Embodiment

[0058] (1) According to the above-described embodiment, the servers **40** transmit the request DM for continuing the travel of the first group GR**1** to the traffic rule display device **30**. Therefore, when the first group GR**1** and the second group GR**2** travel in range in accordance with the indication of the traffic rule display device **30**, the servers **40** can realize that the first group GR**1** and the second group GR**2** continue to travel in a group GR having a higher influence degree IF. Therefore, with respect to the traffic network in which the plurality of vehicles **20** travel, the servers **40** can prevent the impact on the traffic network from becoming large due to the group GR having a higher influence degree IF being stopped.

[0059] (2) According to the above-described embodiment, in the calculation of the influence degree IF, the servers **40** calculate the influence degree IF to be larger as the number NM of the vehicles **20** constituting the group GR increases. Therefore, the servers **40** can easily continue traveling in the group GR of the first group GR**1** and the second group GR**2** in which NM of vehicles **20** constituting the group GR is large.

[0060] (3) According to the above-described embodiment, in the calculation of the influence degree IF, the servers **40** calculate the influence degree IF to be larger as the average speed AV of the vehicles **20** constituting the group GR is larger. Therefore, the servers **40** can easily continue traveling in the group GR of the first group GR**1** and the second group GR**2** in which the average speed AV of the vehicles **20** constituting the group GR is large.

[0061] (4) According to the above-described embodiment, in the calculation of the influence degree IF, the servers **40** calculate the influence degree IF larger than when the emergency vehicle EV is not included in the group GR when the emergency vehicle EV is included. Therefore, it is easy for the servers **40** to continue the traveling of the emergency vehicle EV.

[0062] (5) According to the above-described embodiment, in the calculation of the influence degree IF, the servers **40** calculate the influence degree IF larger when the large-sized vehicle LV is included in the group GR than when the large-sized vehicle LV is not included. Therefore, it is easy for the servers **40** to continue the traveling of the large-sized vehicles LV.

OTHER EMBODIMENTS

[0063] The present embodiment can be realized with the following modifications. The present embodiment and the following modifications can be combined with each other within a technically consistent range to be realized.

[0064] The method of calculating the influence degree IF is not limited to the embodiment described above. For example, when calculating the influence degree IF, CPU **61** may calculate the influence degree IF of the group GR including the emergency vehicle EV to be larger than the influence degree IF of the group GR not including the emergency vehicle EV. Further, for example, CPU **61** may calculate the influence degree IF to be larger when CPU **61** calculates the influence degree IF as the mean of the sizes of the vehicles **20** constituting the group GR is larger. That is, CPU **61** may calculate the influence degree IF regardless of a part or all of the number NM of the vehicle **20**, the average speed AV of the vehicle **20**, the presence or absence of the emergency vehicle EV, and the presence or absence of the large-sized vehicle LV.

[0065] The method of determining the group GR is not limited to the exemplary embodiment described above. For example, CPU **61** may omit S**21** process and identify the group GR by S**22** process. Further, CPU **61** may omit S**22** process and identify the plurality of vehicles **20** extracted by S**21** process as the group GR. Further, for example, when information indicating the destination of the vehicle **20** is included in the moving body information VI in addition to or in place of S**21** and S**22** processes, CPU **61** may identify the group GR based on the information indicating the destination. Specifically, CPU **61** may identify, as the group GR, the vehicles **20** having the same destination as the destination.

[0066] The method of generating the predicted moving body information FI is not limited to the embodiment described above. For example, the reference time may be set as a time that is later than the acquisition point in time of all the vehicles **20**. Even in this situation, CPU **61** can generate the predicted moving body information FI after acquiring the moving body information VI.

[0067] The moving body information VI is not limited to information acquired from the vehicles **20**. For example, the moving body information VI may be information obtained by the servers **40** from the cameras on the information about the vehicles **20** obtained by the cameras provided in the traffic network.

[0068] The configuration of the display unit **31** of the traffic rule display device **30** is not limited to the example of the above-described embodiment. For example, in addition to the red lamp and the blue lamp, a lamp that permits traveling in each direction of travel of the intersection may be included.

[0069] The traffic rule display device **30** is not limited to a traffic signal. For example, it may be an electric bulletin board that displays traffic rules, or it may be a plurality of road tacks. The traffic rule display device **30** may be any device that displays different traffic rules by changing the display state.

[0070] The display state in which the first group GR**1** can continue traveling in the traffic rule display device **30** is not necessarily a display state in which the second group GR**2** cannot continue traveling. As in the above-described embodiment, the server **40** causes the first group GR**1** to be in a displayed state in which the travel can be continued when the first group GR**1** or the second group GR**2** is in a state in which the travel can be continued. At the same time, the driving of the second group GR**2** may not be continued.

[0071] The servers **40** may determine whether or not to execute S**53** and S**62** processes by referring to the display status of the traffic rule display device **30** of the predicted moving body information FI. For example, when S**53** process is executed, S**53** process may be executed only when the display state of the traffic rule display device **30** in the predicted moving body information FI is not a display state in which the traveling of the first group GR**1** can be continued.

[0072] The information processing device **60** may be configured as a circuitry including one or more processors that execute various processes in accordance with a computer program (software). Note that the information processing device **60** may be configured as a circuit including one or more dedicated hardware circuits such as an application-specific integrated circuit (ASIC) that executes at least some of the various processes, or a combination thereof. The processor includes a CPU and a memory such as a random access memory (RAM) and a ROM. The memory stores a program code or an instruction configured to execute the CPU to perform processes. The memory, that is, the computer-readable medium includes any available media that can be accessed by a general purpose or special purpose computer.

What is claimed is:

1. A server that transmits a control signal based on predicted moving body information, that is generated based on moving body information that is information of a vehicle in a real world, following a point in time of acquiring the moving body information, to a traffic rule display device that displays different traffic rules by changing a display state, the server executing

identifying a first group, and a second group different from the first group, as groups in which a plurality of the vehicles travels in a group, based on the predicted moving body information,

calculating, for each of the groups identified, an influence degree, indicating a degree of influence imparted on a traffic network when the group that is an object of calculation is stopped,

calculating, when the first group and the second group are located within a vicinity range including a range of traveling in accordance with a display of the traffic rule display device, the display state under which the group of the first group and the second group of which the influence degree is higher can continue traveling, and

transmitting a request, that is the display state that is calculated, to the traffic rule display device.

2. The server according to claim **1**, wherein, when calculating the influence degree, the greater a count of the vehicles making up the group that is the object of calculation is, the greater the influence degree is calculated to be.

3. The server according to claim **1**, wherein, when calculating the influence degree, the higher an average speed of the vehicles making up the group that is the object of calculation is, the greater the influence degree is calculated to be.

4. The server according to claim **1**, wherein, when calculating the influence degree, the influence degree is calculated to be greater when the group that is the object of calculation includes an emergency vehicle, than when not including the emergency vehicle.

5. The server according to claim **1**, wherein, when calculating the influence degree, the influence degree is calculated to be greater when the group that is the object of calculation includes a large-sized vehicle, than when not including the large-sized vehicle.

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